

Overview of Experiment

General Instructions

Welcome to this experiment!

For participating in today's experiment, **you will receive a participation fee of €6.00**. You may earn additional money depending on your decisions. Hence, please read the following instructions carefully.

There are three parts to today's experiment: Part A, Part B, and Part C. You will be asked to complete a questionnaire after Part C. You will receive detailed instructions for each part before you participate in them. Note that your decisions in one part will not affect your earnings from the other parts.

You will be paid for the decisions you make in **one** of the three parts, chosen at random by the computer. This implies that you should carefully consider all decisions you make in each part as they may determine your earnings.

You will be informed of your payment at the end of the experiment. The payment will be done in cash at the end of the experiment.

Finally, please note that this is an economics experiment, which **does not employ deception**.

If you have a question, please raise your hand and an experimenter will come to you. Your question will then be answered in private. All the decisions you make in this experiment will be anonymous.

CONTINUE

Random Numbers in the Experiment

In this experiment, there are a number of stages that will involve random draws of numbers to determine outcomes and payments.

To ensure transparency in the payment process, we have pre-determined all random draws prior to the experiment. The pre-determined draws are stored in the following Excel spreadsheet.

Please feel free to download the file for your record. The password to open this file will be provided to you at the end of the experiment.

To determine the outcomes and payments for **you** in this experiment, we will use the draws corresponding to **ID number 734** in the spreadsheet. You may wish to write this number down.

You may download the file by clicking [here](#).

CONTINUE

Part A (Belief Elicitation) – Instructions

Brief Instructions

Instructions for Part A

We will now begin Part A. Please proceed to read the instructions over the next few pages.

We recommend you review the instructions carefully before proceeding.

Next

Instructions for Part A

Part A consists of six rounds of an individual decision-making task. At the end of the experiment, if you are paid for Part A, then the computer will randomly pick **one of the six rounds** for payment.

In each round, your task is to predict the chances that an "event" will occur.

For example: What is the chance that a random card drawn from a full deck of standard playing cards is a heart?
We want you to predict this chance from 0 to 100%.

You will want to think about your answer carefully. This is because, based on the prediction you give, we will use a payment procedure that rewards you with up to 100 lottery tickets to earn a prize of 15 euros.

Each lottery ticket corresponds to a 1% chance to win the prize of 15 euros. So, the more lottery tickets you have, the higher your chances will be to win.

The number of lottery tickets you will receive is determined by the following formulas:

$$\text{Number of lottery tickets if Event occurs} = \left[1 - \left(\frac{100 - \text{your prediction}}{100} \right)^2 \right] \times 100$$

$$\text{Number of lottery tickets if Event does not occur} = \left[1 - \left(\frac{\text{your prediction}}{100} \right)^2 \right] \times 100$$

In summary, you will receive more lottery tickets on average if your prediction of the Event is more accurate.

CONTINUE

End of Instructions

Before we proceed, please answer the following question:

Q1: How certain are you about your understanding of how you will be paid for your predictions?

☐ 0% ☐ 5% ☐ 10% ☐ 15% ☐ 20% ☐ 25% ☐ 30% ☐ 35% ☐ 40% ☐ 45% ☐ 50% ☐ 55% ☐ 60% ☐ 65% ☐ 70% ☐ 75% ☐ 80% ☐ 85% ☐ 90% ☐ 95% ☐ 100%

Very uncertain **Completely certain**

SUBMIT

End of Instructions

Before we proceed, please answer the following question:

Q2: How certain are you about how the number of lottery tickets you will receive is calculated?

☐ 0% ☐ 5% ☐ 10% ☐ 15% ☐ 20% ☐ 25% ☐ 30% ☐ 35% ☐ 40% ☐ 45% ☐ 50% ☐ 55% ☐ 60% ☐ 65% ☐ 70% ☐ 75% ☐ 80% ☐ 85% ☐ 90% ☐ 95% ☐ 100%

Very uncertain **Completely certain**

SUBMIT

Detailed Instructions

Instructions for Part A

We will now begin Part A. Please proceed to read the instructions over the next few pages.

As we progress, you will be asked to answer a set of comprehension questions.

If you make too many failed attempts on too many questions, then we may remove you from the experiment, and you will forgo the payments from the rest of the experiment.

Hence, we recommend you review the instructions carefully before proceeding.

Next

Instructions for Part A

Part A consists of six rounds of an individual decision-making task. At the end of the experiment, if you are paid for Part A, then the computer will randomly pick **one of the six rounds** for payment.

In each round, your task is to predict the chances that an "event" will occur.

For example: What is the chance that a random card drawn from a full deck of standard playing cards is a heart?
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$$\text{Number of lottery tickets if Event occurs} = \left[1 - \left(\frac{100 - \text{your prediction}}{100} \right)^2 \right] \times 100$$

$$\text{Number of lottery tickets if Event does not occur} = \left[1 - \left(\frac{\text{your prediction}}{100} \right)^2 \right] \times 100$$

In summary, you will receive more lottery tickets on average if your prediction of the Event is more accurate.

We will now provide a detailed explanation of the payment procedure that we introduced on this page by using three short **videos**.

Please watch the videos over the next few pages carefully.

CONTINUE

Instructions for Part A (Video 1)

Please watch the videos over the next few pages carefully. The videos are set up such that you must watch each video entirely. You will be asked to answer a set of comprehension questions after watching each video to make sure you have understood the instructions.

Please watch Part 1 of the video below.



Note: To read the transcript of the video, click this link (you will remain on this page): [Read Video Script](#)

By adjusting your prediction on the slider below, you will see how many lottery tickets you would receive.

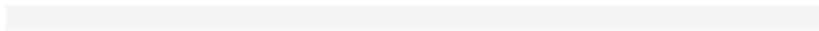
When answering the questions below, imagine that a **heart** is drawn



Please adjust the slider.



Please adjust the slider.



Comprehension Questions

Please answer the following questions.

- (1) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **0%** chance, and a **heart turned out to be drawn**?

- (2) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **25%** chance, and a **heart turned out to be drawn**?

- (3) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **75%** chance, and a **heart turned out to be drawn**?

- (4) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **100%** chance, and a **heart turned out to be drawn**?

CHECK ANSWERS

Instructions for Part A (Video 2)

Please watch Part 2 of the video below.



Note: To read the transcript of the video, click this link (you will remain on this page): [Read Video Script](#)

By adjusting your prediction on the slider below, you will see how many lottery tickets you would receive.

When answering the questions below, imagine that a **spade, club, or diamond** is drawn



Please adjust the slider.



Please adjust the slider.

Comprehension Questions

Please answer the following questions.

- (1) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **0%** chance, and a **spade, club, or diamond** turned out to be drawn?

- (2) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **25%** chance, and a **spade, club, or diamond** turned out to be drawn?

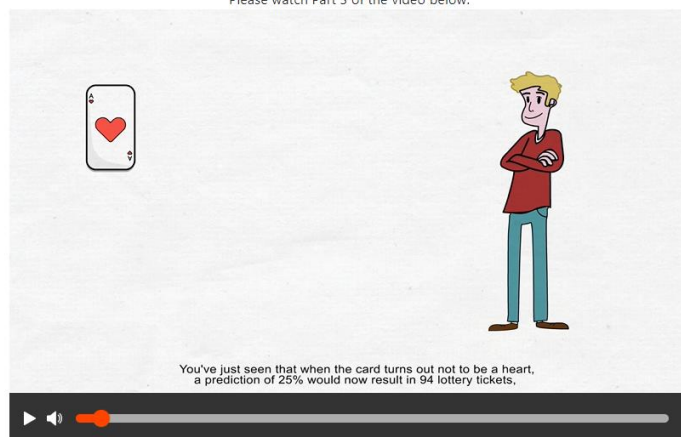
- (3) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **75%** chance, and a **spade, club, or diamond** turned out to be drawn?

- (4) What is your percentage chance (whole number) of receiving €15 if you predict that a heart would be drawn with a **100%** chance, and a **spade, club, or diamond** turned out to be drawn?

[CHECK ANSWERS](#)

Instructions for Part A (Video 3)

Please watch Part 3 of the video below.



Note: To read the transcript of the video, click this link (you will remain on this page): [Read Video Script](#)

Suppose you know that the chance a randomly drawn card will be a heart is **25%**

Drag the slider to enter your prediction.

As you adjust the slider, we will show you how many lottery tickets you would receive **on average**, assuming the correct estimate that the chance of a heart is **25%**.



Please adjust the slider.



Please adjust the slider.

Comprehension Questions

Suppose you know that the chance a randomly drawn card will be a heart is **25%**. Please answer the following questions.

- (1) What is your percentage chance (to 2 decimal places) of receiving €15 **on average** if you predict that a heart would be drawn with a **0%** chance?

- (2) What is your percentage chance (to 2 decimal places) of receiving €15 **on average** if you predict that a heart would be drawn with a **25%** chance?

- (3) What is your percentage chance (to 2 decimal places) of receiving €15 **on average** if you predict that a heart would be drawn with a **75%** chance?

- (4) What is your percentage chance (to 2 decimal places) of receiving €15 **on average** if you predict that a heart would be drawn with a **100%** chance?

- (5) What is the **prediction** that gives you the **highest** chance of receiving €15 on average?

CHECK ANSWERS

End of Instructions

You have answered all the comprehension questions correctly.

Before we proceed, please answer the following question:

Q1: How certain are you about how the number of lottery tickets you will receive is calculated?

☐

0%

☐

5%

☐

10%

☐

15%

☐

20%

☐

25%

☐

30%

☐

35%

☐

40%

☐

45%

☐

50%

☐

55%

☐

60%

☐

65%

☐

70%

☐

75%

☐

80%

☐

85%

☐

90%

☐

95%

☐

100%

Very uncertain

Completely certain

SUBMIT

End of Instructions

You have answered all the comprehension questions correctly.

Before we proceed, please answer the following question:

Q2: How certain are you about your understanding of how you will be paid for your predictions?

☐

0%

☐

5%

☐

10%

☐

15%

☐

20%

☐

25%

☐

30%

☐

35%

☐

40%

☐

45%

☐

50%

☐

55%

☐

60%

☐

65%

☐

70%

☐

75%

☐

80%

☐

85%

☐

90%

☐

95%

☐

100%

Very uncertain

Completely certain

SUBMIT

All Treatments

Understanding Check Questions

Please answer the following question. You will receive €0.05 if your answer is correct.

What statement best reflects the payment procedure?

- ☐ My payment depends both on the outcome of the event and my prediction.
- ☐ My payment depends only on my prediction, and not on the outcome of the event.

SUBMIT ANSWER

Understanding Check Questions

Please answer the following question. You will receive €0.05 if your answer is correct.

Suppose you know for a fact that an Event A is going to occur.

And you know for a fact that Event B is NOT going to occur.

What are the two predictions that will result in the highest number of lottery tickets?

- ☐ Predictions - Event A: 50%, Event B: 50%
- ☐ Predictions - Event A: 75%, Event B: 25%
- ☐ Predictions - Event A: 100%, Event B: 0%
- ☐ Predictions - Event A: 0%, Event B: 100%

SUBMIT ANSWER

Understanding Check Questions

Please answer the following question. You will receive €0.05 if your answer is correct.

Suppose you are predicting an event, like the probability that a fair coin toss comes up Heads.

It turns out that the coin is flipped and the result is Heads.

Given that the coin was Heads, what type of prediction would have resulted in the most lottery tickets?

- ☐ A high prediction
- ☐ A prediction in the middle
- ☐ A low prediction

SUBMIT ANSWER

Understanding Check Questions

Please answer the following question. You will receive €0.05 if your answer is correct.

Which of the following statements best reflects the payment procedure?

- ☐ The payment procedure is designed so that I should expect to earn the most money ON AVERAGE if I enter my best prediction of a given event.
- ☐ The payment procedure is designed so that I should expect to earn the most money ON AVERAGE if I enter a prediction of 0% for unlikely events, and 100% for likely events.
- ☐ The payment procedure is designed so that I should expect to earn the most money ON AVERAGE if I enter a prediction of 50%.

SUBMIT ANSWER

Understanding Check Questions

Please answer the following question. You will receive €0.05 if your answer is correct.

Suppose that you made a prediction that Event C would happen with a 0% chance.

Then, Event C did NOT happen.

How many lottery tickets would you receive?

SUBMIT ANSWER

Instructions for Part A (Page 1)

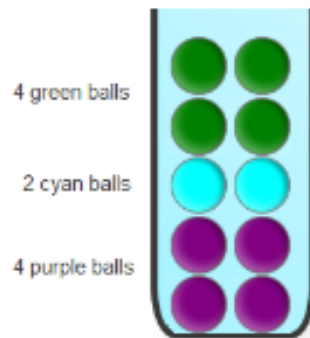
We are now ready to begin Part A.

Instead of predicting the card drawn from a set of playing cards, **you will make predictions pertaining to the color of a ball drawn from an urn.**

Each round, the computer will present you with an **urn containing 10 balls**. You will always be told how many possible colors there can be in the urn.

For example, **in the urn below** there can be **up to 5 colors**: Green, Gray, Cyan, Purple, Pink. These are the only possible colors.

The exact composition turns out to be:



Example 1:

PREDICTED EVENT: The randomly drawn ball is **Purple**.

In example 1, your task is to predict the chances that a randomly drawn ball from this urn is Purple. You may note that **there are 4 purple balls**.

Example 2:

PREDICTED EVENT: The randomly drawn ball is **Pink or Gray**.

In example 2, your task is to predict the chances that a randomly drawn ball from this urn is either Pink or Gray. You may note that **there are 0 Pink or Gray balls**.

Your payment based on your prediction in Part A will follow the **payment procedure** we just explained.

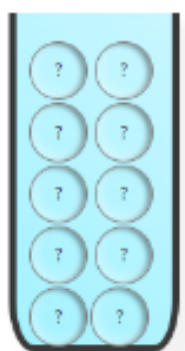
Each lottery ticket (up to a maximum of 100 lottery tickets) corresponds to a 1% chance to win the prize of 15 euros.

Next

Instructions for Part A (Page 2)

Each round the urn will be different. In some cases, you will not know the exact urn composition.

For example, in the urn below there can be up to 5 colors: Blue, Orange, Teal, Red, Yellow. These are the only possible colors.



Example:

PREDICTED EVENT: The randomly drawn ball is Red.

In this example your task is to predict the chances that a randomly drawn ball from this urn is Red. You do not know how many of the balls are Red. You only know that there are up to 5 colors.

Your payment based on your prediction in Part A will follow the **payment procedure** we just explained.

Each lottery ticket (up to a maximum of 100 lottery tickets) corresponds to a 1% chance to win the prize of 15 euros.

Please answer the following comprehension question.

In the urn above, what do you know about the number of Red balls?

- ☐ There is a minimum of 1 red ball.
- ☐ There can be any number of red balls, from 0 to 10.
- ☐ There is a maximum of 6 red balls, since there must be 4 other colors.

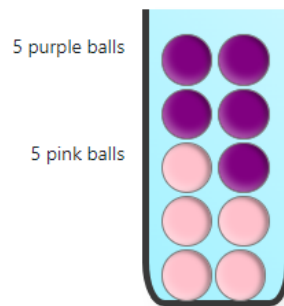
Previous

CHECK ANSWER

Part A (Belief Elicitation) – Main Task

Part A (Round 4)

In the urn below there can be **up to 2 colors**, and the urn's composition is shown in the figure below.



PREDICTED EVENT: The randomly drawn ball **is Pink**.

What is your prediction (out of 100) that a randomly drawn ball is Pink?

You predict a % chance that ball is
Purple.

You predict a % chance that ball is
Pink.

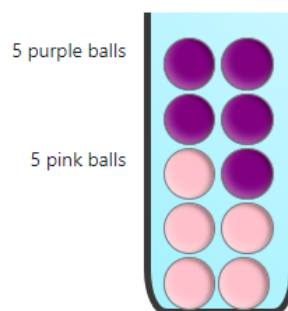
Reminder: Your chance of receiving €15 will depend *both* on your prediction and which ball is drawn from this urn, according to the **payment procedure**.

Click this link to see the instructions again (you will remain on this page): [Review Payment Procedure](#)

SUBMIT

Part A (Round 4)

In the urn below there can be **up to 2 colors**, and the urn's composition is shown in the figure below.



You are asked to predict the chance that ball is Pink.

Your prediction is 64%.

Do you want to submit your decision?

GO BACK

SUBMIT DECISION

Part A (Round 4)

Your prediction was **64%**.

Given the **payment procedure**, how **certain** are you that the **optimal answer** is between 63% and 65%?

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
0%	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	100%
Very uncertain											Completely certain									

SUBMIT

Post-Experimental Questionnaire

Questionnaire

We have completed the main tasks in the experiment.

Before concluding the experiment and showing your payment summary, please proceed to answer the questionnaire.

Please be reminded that all your responses will remain anonymous.

Next

Questionnaire

In **Round 5** of Part A, you faced the following urn, which has 8 Green, Gray, Yellow, and White balls (that is, 80% of the urn was either Green, Gray, Yellow, or White).



Your prediction that a randomly drawn ball is either Green, Gray, Yellow, or White was 77%.

Why did you choose 77% (and not 80%)?

- ☐ I chose **randomly**.
- ☐ I **did not understand** that 80% of the urn was Green, Gray, Yellow, or White.
- ☐ I **did not believe** that 80% of the urn was Green, Gray, Yellow, or White.
- ☐ I **believed** that 80% of the urn was Green, Gray, Yellow, or White, but I chose a different answer **so that I could earn more money according to the payment procedure**.
- ☐ Other

If you select Other, please explain below:

Next

Questionnaire - Quiz Question

Please answer the following quiz question. You will receive €0.05 if your answer is correct.

If Alex can drink one barrel of water in 6 days, and Christopher can drink one barrel of water in 12 days,
how many days would it take them to drink one barrel of water together?

Next

Questionnaire - Quiz Question

Please answer the following quiz question. You will receive €0.05 if your answer is correct.

Ben received both the 15th highest and the 15th lowest mark in the class.

How many students are in the class?

Next

Questionnaire - Quiz Question

Please answer the following quiz question. You will receive €0.05 if your answer is correct.

A man buys a pig for €60, sells it for €70, buys it back for €80, and sells it finally for €90.

How much (in €) has he made?

Next

Questionnaire - Quiz Question

Please answer the following quiz question. You will receive €0.05 if your answer is correct.

Fraser decided to invest €8,000 in the stock market one day early in 2008.

Six months after he invested, on July 17, the stocks he had purchased were down 50%.

Fortunately for Fraser, from July 17 to October 17, the stocks he had purchased went up 75%.

At this point, Fraser has:

- ☐ broken even in the stock market.
- ☐ is ahead of where he began.
- ☐ has lost money.

Next

Questionnaire

Did you find it easy or difficult to understand the instructions?

How much effort did it take you to understand the instructions?

Were there any specific parts of the experiment that were not clear to you?

If yes, please explain briefly.

Was there anything about the videos that did not work for you or could be improved? Please explain.

Next

Questionnaire

What is your field of study?

If you select Other, please state below:

How old are you?

Gender: How do you identify?

Do you have any form of color blindness?

☐ No

☐ Yes

Next